

Project 4 - Create a VPC with Public and Private Subnets Report

Introduction

Cloud computing requires secure and scalable network infrastructure to host applications and services. Amazon Virtual Private Cloud (VPC) is a networking service provided by AWS that enables users to create isolated virtual networks within the AWS Cloud. A VPC allows users to define IP address ranges, create subnets, configure route tables, and control network connectivity.

This project focuses on creating a custom VPC with one public subnet and one private subnet. The public subnet is configured for internet access using an Internet Gateway, while the private subnet remains isolated to enhance security. This project demonstrates the fundamental concepts of AWS networking and network segmentation.

Objective

The main objective of this project is to:

- ✧ Create a custom Amazon VPC
- ✧ Create one public subnet
- ✧ Create one private subnet
- ✧ Create and attach an Internet Gateway
- ✧ Configure route tables
- ✧ Associate the public subnet with a route table
- ✧ Verify VPC network architecture
- ✧ Understand AWS networking fundamentals

Overview of Amazon VPC

Amazon Virtual Private Cloud (VPC) enables users to create a logically isolated network environment within AWS. It provides complete control over IP addressing, subnet creation, routing, and internet connectivity.

Key Concepts

- ❖ **VPC:** A logically isolated virtual network in AWS
- ❖ **Subnet:** A segment of a VPC IP address range where resources can be deployed
- ❖ **Public Subnet:** A subnet that has access to the internet through an Internet Gateway
- ❖ **Private Subnet:** A subnet that does not have direct internet access
- ❖ **Internet Gateway (IGW):** A gateway that enables communication between a VPC and the internet
- ❖ **Route Table:** A set of routing rules that determine where network traffic is directed
- ❖ **CIDR Block:** A range of IP addresses assigned to a VPC or subnet

Methodology

Creating the VPC

A custom VPC named **Project4-VPC** was created using the AWS Management Console. The IPv4 CIDR block **10.0.0.0/16** was assigned to provide a private IP address range for the network.

Creating Public and Private Subnets

Two subnets were created within the VPC:

- ❖ Public Subnet: **10.0.1.0/24**
- ❖ Private Subnet: **10.0.2.0/24**

The public subnet was configured to automatically assign public IPv4 addresses to resources launched within it.

Creating and Attaching an Internet Gateway

An Internet Gateway named **Project4-IGW** was created and attached to the VPC. This gateway enables resources in the public subnet to communicate with the internet.

Configuring the Route Table

A route table named **Public-RT** was created and configured with the following route

- ❖ Destination: **0.0.0.0/0**
- ❖ Target: **Project4-IGW**

This route directs internet-bound traffic through the Internet Gateway.

Associating the Public Subnet

The Public-Subnet was explicitly associated with the Public-RT route table. This enabled internet connectivity for resources deployed in the public subnet.

The Private-Subnet was not associated with the public route table and therefore remained isolated from direct internet access.

Verifying the Architecture

The AWS VPC Resource Map was used to verify:

- ❖ VPC creation
- ❖ Subnet configuration
- ❖ Route table association
- ❖ Internet Gateway attachment
- ❖ Public and private network segmentation

Results

- ❖ Successfully created a custom Amazon VPC
- ❖ Created public and private subnets

- ❖ Configured auto-assignment of public IP addresses for the public subnet
- ❖ Created and attached an Internet Gateway
- ❖ Configured a route table for internet access
- ❖ Associated the public subnet with the route table
- ❖ Verified VPC architecture using the Resource Map
- ❖ Successfully implemented a basic AWS network infrastructure

Key Learnings

- ❖ Amazon VPC provides an isolated networking environment in AWS
- ❖ Public subnets require a route to an Internet Gateway for internet connectivity
- ❖ Route tables control the flow of network traffic within a VPC
- ❖ Internet Gateways enable communication between AWS resources and the internet
- ❖ Private subnets improve security by restricting direct internet access
- ❖ Proper network segmentation is essential for secure cloud architecture

Advantages of Using VPC with Public and Private Subnets

- ❖ Improved network security
- ❖ Better traffic management
- ❖ Logical separation of resources
- ❖ Scalable cloud networking architecture
- ❖ Supports secure application deployment
- ❖ Integrates seamlessly with other AWS services

Limitations

- ❖ Requires proper route table configuration
- ❖ Misconfigured security settings may affect connectivity
- ❖ Additional components such as NAT Gateway are required for private subnet internet access
- ❖ Managing multiple subnets and routes can increase complexity
- ❖ Network design must be planned carefully for large-scale deployments

Conclusion

This project provided practical experience in creating and managing AWS networking resources using Amazon VPC. It demonstrated how to create a custom VPC, configure public and private subnets, attach an Internet Gateway, and manage routing through route tables. The project enhanced understanding of cloud networking, network segmentation, and secure infrastructure design within AWS.